Bloodhound Auxiliary Sensors

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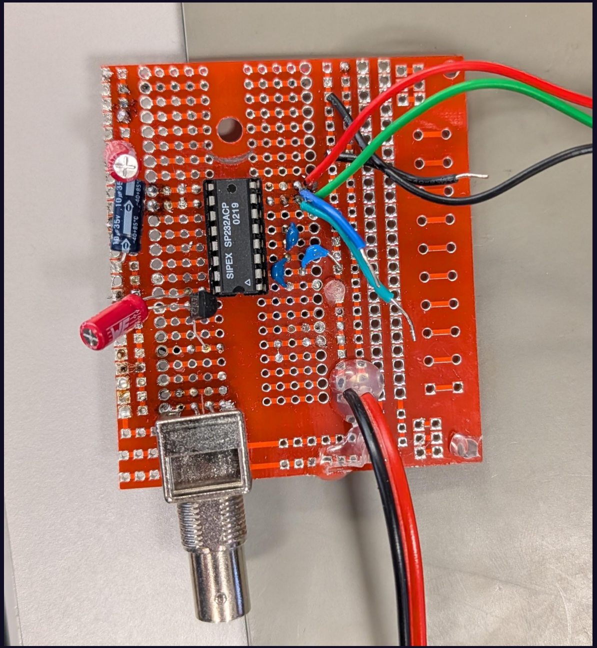
Background/Goals

- Camera project started in 2023 by Carson Bogart and Dane Davids
- LiDAR started Spring 2026

- Computer attached VB program ——---------------------> discrete universal microcontroller control
- Multiple cameras
- Integrate other sensors
- Refine camera control

- Figure out how to utilize 360 LiDAR info

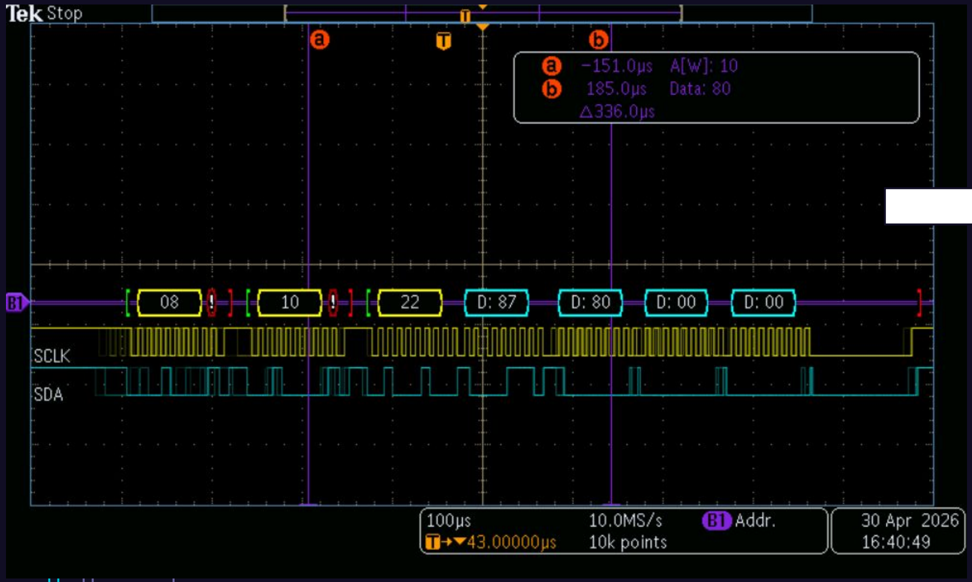


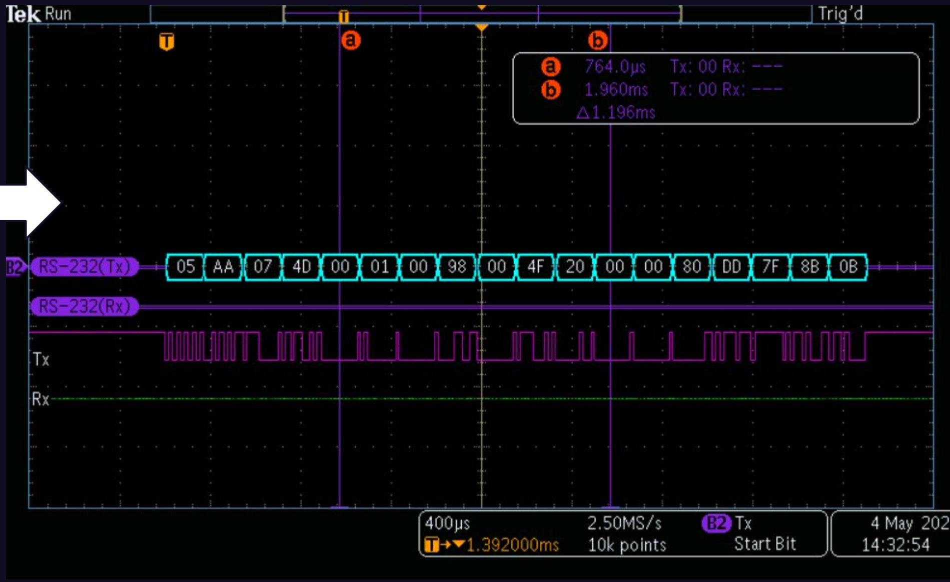
System

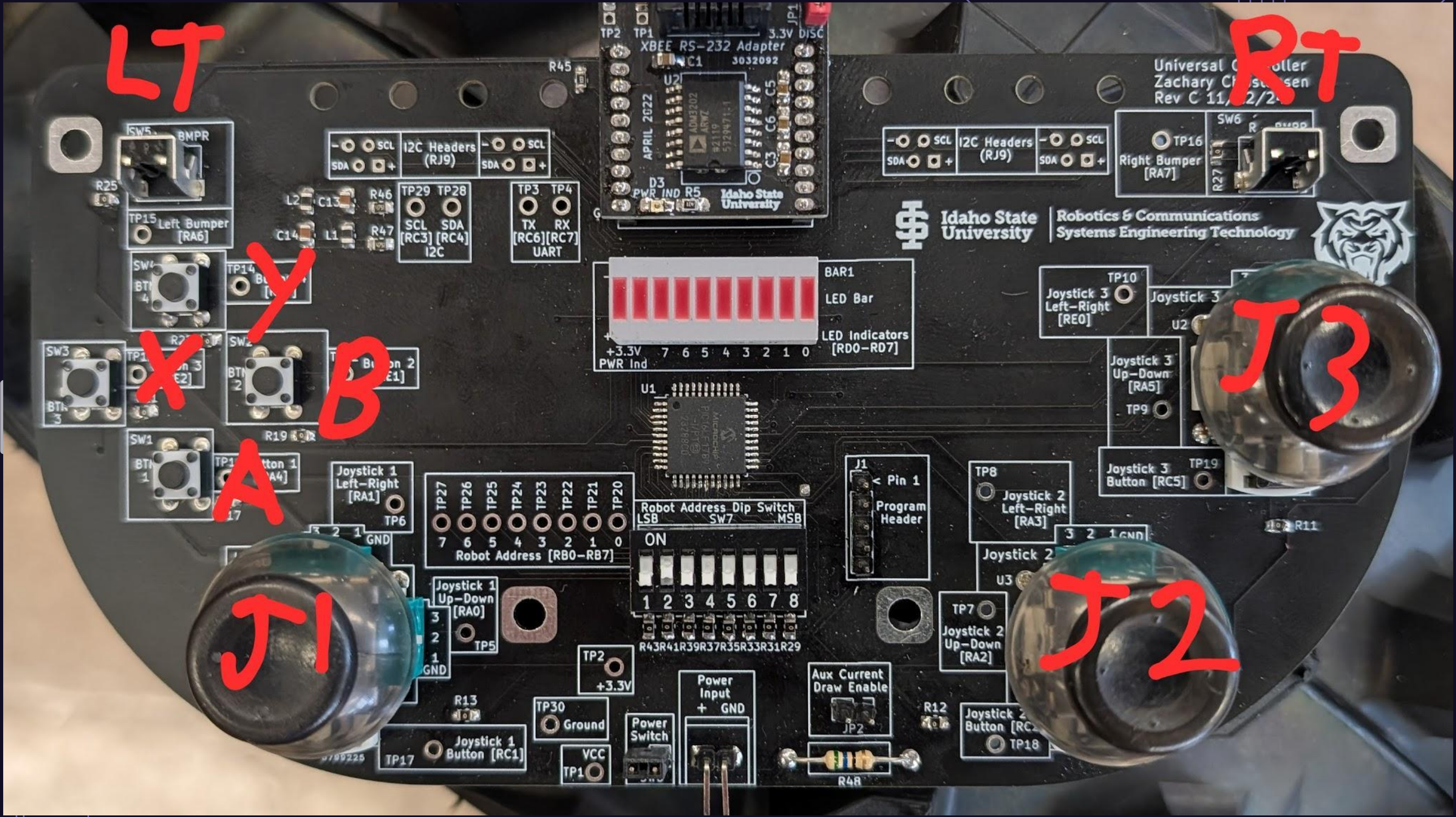
- Bloodhound Search and Rescue robot
- GPS
- LiDAR
- Camera Control

▶ Camera Control

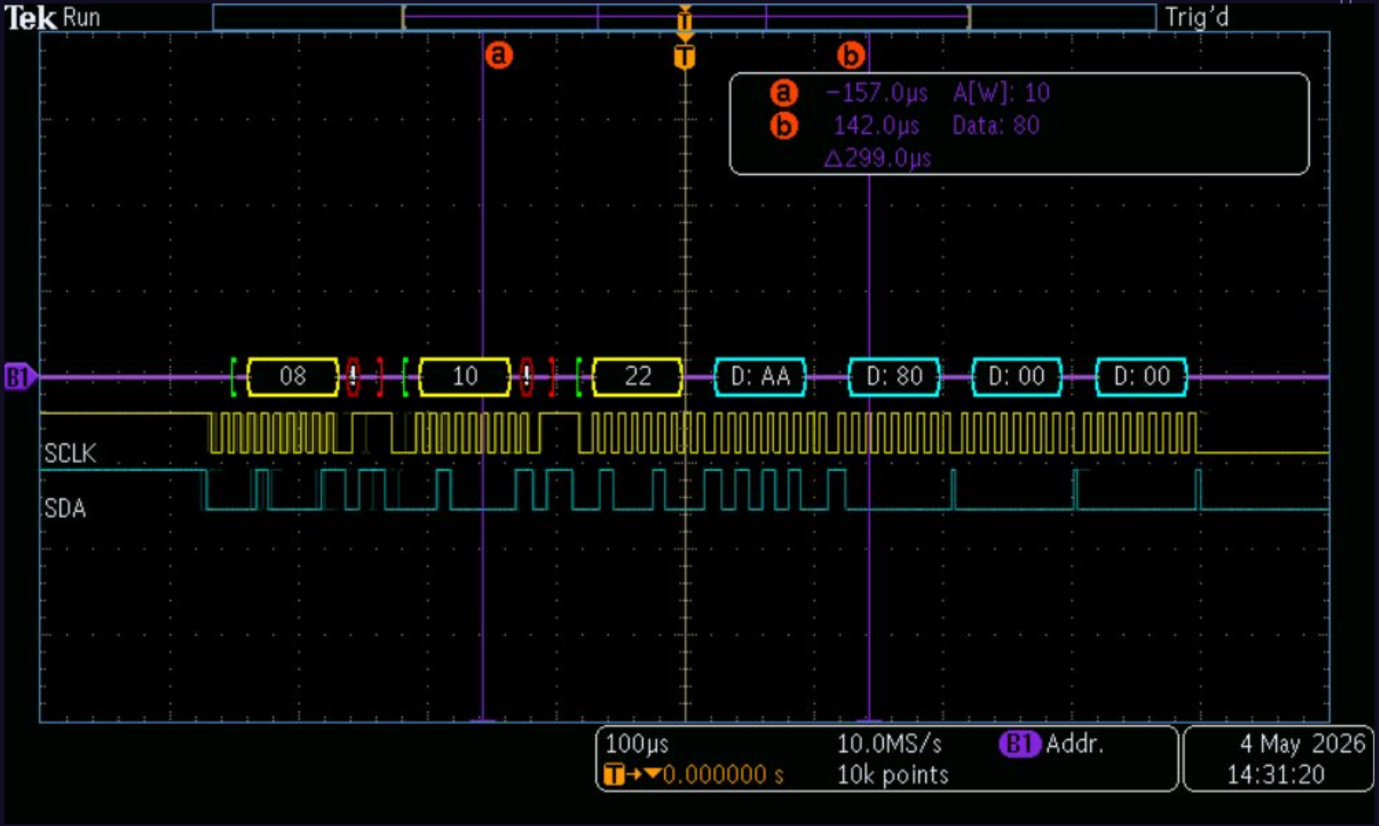
- Universal application with Universal Controller
- Up to 4 cameras
- Control Hood Tech cameras





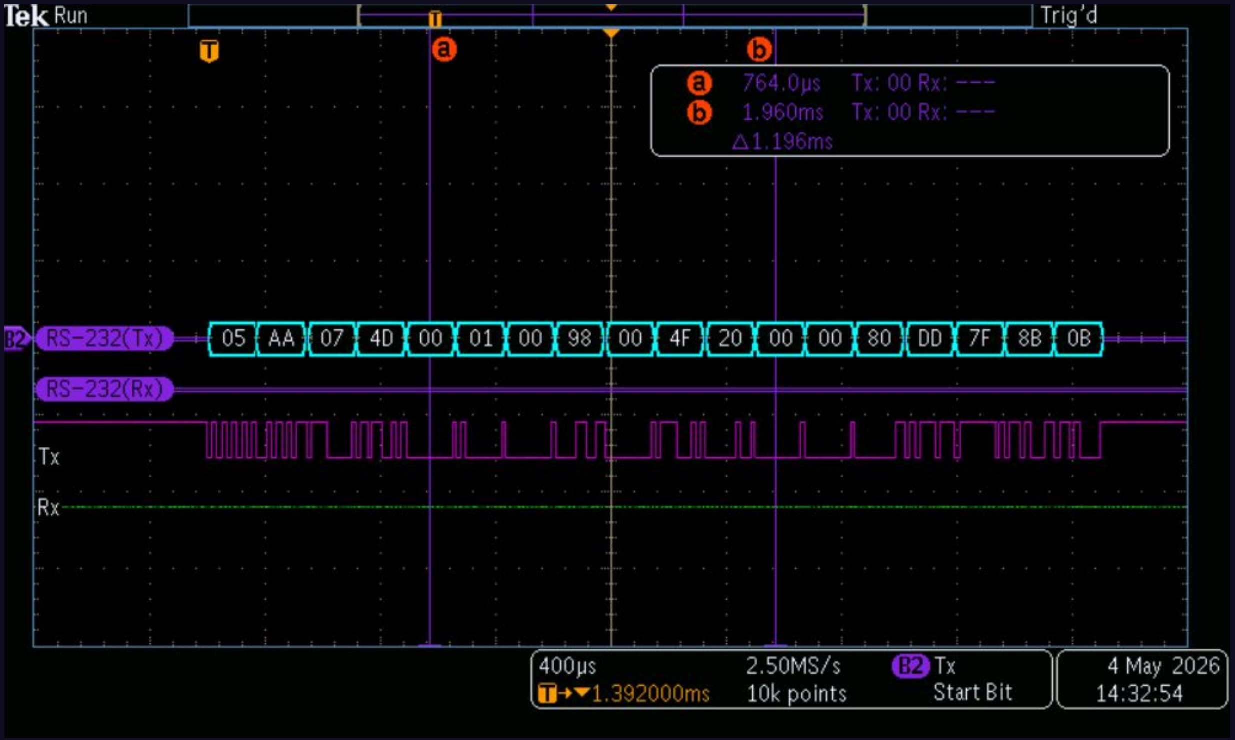

LT
RT
Y
X
B
A
J1
J2
J3
XBEE RS-232 Adapter
Universal Controller
Rev C
Left Bumper [RA6]
Right Bumper [RA7]
I2C Headers (RJ9)
Idaho State University
Robotics & Communications Systems Engineering Technology
LED Bar
LED Indicators [RD0-RD7]
+3.3V PWR Ind
Robot Address [RB0-RB7]
Robot Address Dip Switch
LSB SW7 MSB
ON
Program Header
Joystick 1 Left-Right [RA1]
Joystick 1 Up-Down [RA0]
Joystick 1 Button [RC1]
Joystick 2 Left-Right [RA3]
Joystick 2 Up-Down [RA2]
Joystick 3 Left-Right [RE0]
Joystick 3 Up-Down [RA5]
Joystick 3 Button [RC5]
Power Input
Power Switch
Aux Current Draw Enable
TP30 Ground
TX [RC6] RX [RC7] UART
SCL [RC3] SDA [RC4] I2C

- Address
- Joy 3 Y POS
- Joy 3 X POS
- TABXY
- Joy 3 Button



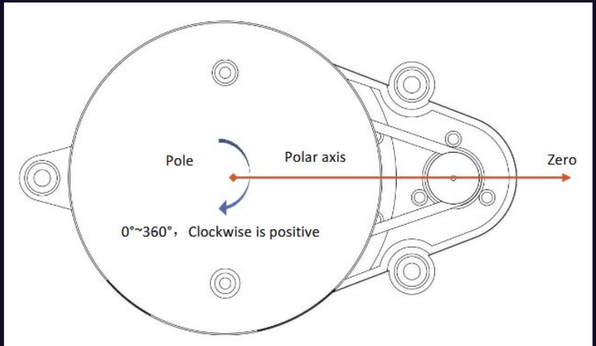
Command	Bytes Sent
Up .2 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, DD, 7F, 8B, 0B
Up 1 degree	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, 51, 7F, 4B, 6F
Up 5 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, 97, 7C, EA, 7C
Down .2 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, 23, 80, AB, 0B
Down 1 degree	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, AF, 80, 6B, 6F
Down 5 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, 69, 83, CA, 7C
Left .2 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, DD, 7F, 00, 80, C7, 18
Left 1 degree	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 51, 7F, 00, 80, 57, 32
Left 5 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 97, 7C, 00, 80, DF, F3
Right .2 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 23, 80, 00, 80, 1F, 19
Right 1 degree	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, AF, 80, 00, 80, 8F, 33
Right 5 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 69, 83, 00, 80, 07, FF
Zoom in	55, AA, 07, 4D, 00, 01, 00, 98, 00, 5A, 01, 00, 00, 00, 01, 00, F3, D8
Zoom out	55, AA, 07, 4D, 00, 01, 00, 98, 00, 5A, FF, 00, 00, 00, 01, 00, 2D, CD
Zoom stop	55, AA, 07, 4D, 00, 01, 00, 98, 00, 5A, 00, 00, 00, 00, 00, 00, B2, DB
Video On	55, AA, 07, 4D, 00, 01, 00, 98, 00, 7C, 00, 00, 00, 00, 00, 00, 70, 9F
Home	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 4A, 19, B8, 8B, 00, 80, 31, 53
Reset Camera	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, FE, 00, 00, 00, 00, 00, 6D, 00
FFC/Adjust	55, AA, 07, 4D, 00, 01, 00, 98, 00, 6D, 01, 03, 00, 00, 00, 00, A0, DB

Command	Bytes Sent
Up .2 degrees	55, AA, 07, 4D, 00, 01, 00, 98, 00, 47, 20, 00, 00, 80, DD, 7F, 8B, 0B



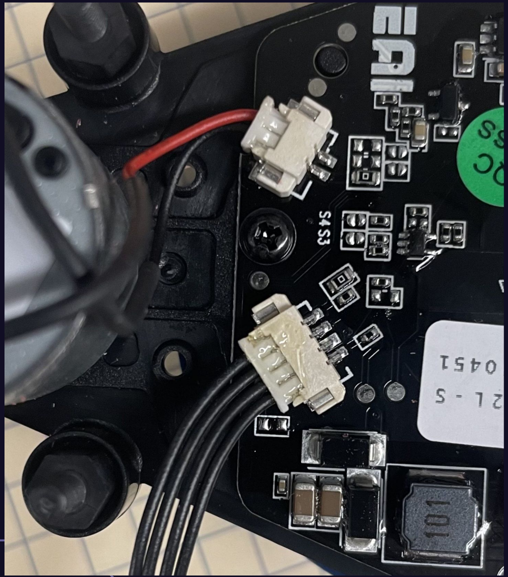
LiDAR - Light Detection and Ranging

- YDLiDAR X2L Module (image)
- Basic function:
  - Rotating assembly emits low-level IR light
  - Received light/signal is timed and converted to digital value
  - Digital value along with angle sector is output
  - Process is repeated quickly for 360 degree peripheral detection
  - Scan angle embedded sent with data



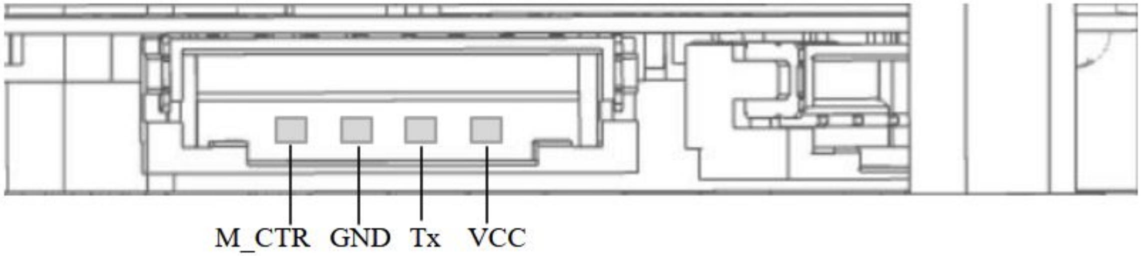
LiDAR Connections

- LiDAR module only OUTPUTS data
- No data input



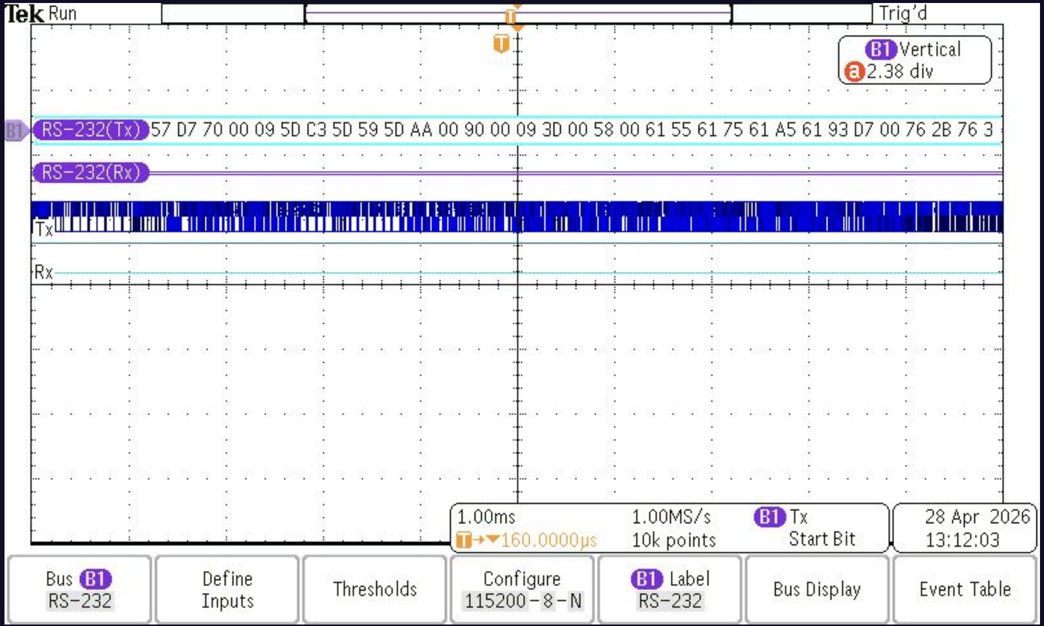
Interface definition

X2 provides a PH1.25-4P female connector with functional interfaces for system power, data communication and motor control.



Reading LiDAR

- Baud rate - 115,200
- 8-Bit character bytes
- No obvious start/stop bits



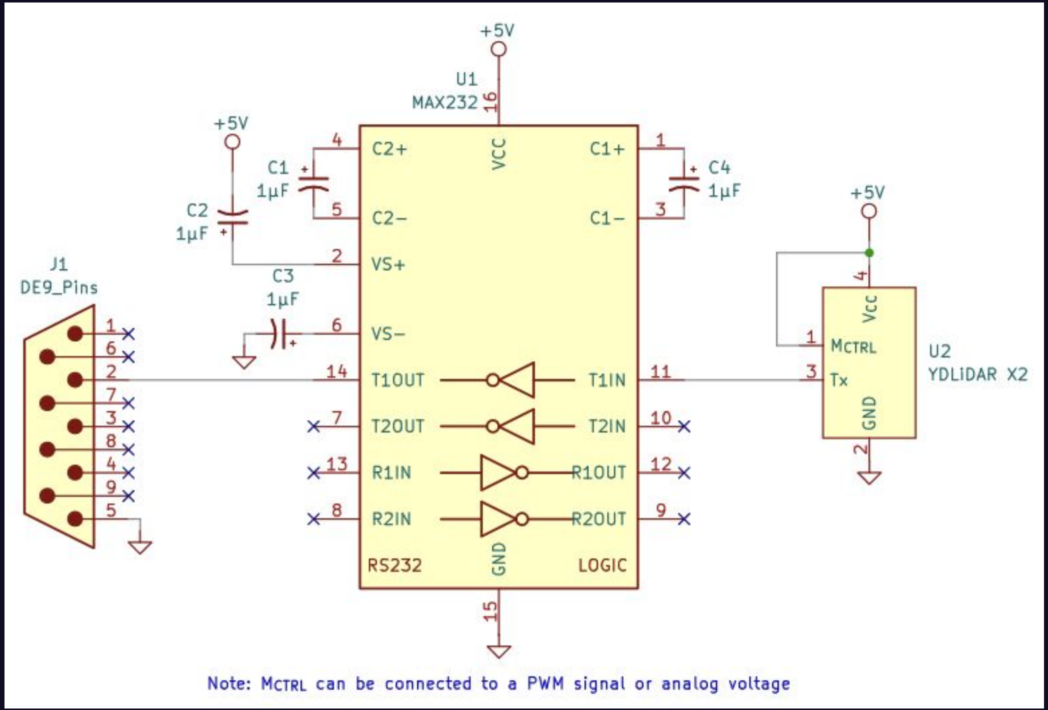
Deciphering LiDAR Packet

- Manufacturer Software terminal Output

```
[2026-04-22 13:16:48][INFO] Start lidar [2026-04-22 13:16:48][DEBUG] [700]??:AA5500286B58D776894B [2026-04-22 13:16:48][DEBUG]
[700]??:F207000800000000000000000BE08C1080000000000520B02730000CA5F0A5CCE58FA5645576658555971 5A9A5BE25C4D5DB95CE6570A5506521A4FD24C523AB64534450000000000009A214D2119220123 [2026-04-22 13:16:48]
[DEBUG] ???????[176.83 ~237.67 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA550028A377159675B5 [2026-04-22 13:16:48][DEBUG]
[700]??:2523B5221522BD22B2237C2400000A33EE313A29B6303C30663120316604F603BE03960376035603360321030D03F902ED02DD02D502C902C102B902B102A902A502A1029D029D029D02A10200008202 [2026-04-22 13:16:48]
[DEBUG] ???????[239.27 ~300.16 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA55001ECD9649AD3270 [2026-04-22 13:16:48][DEBUG]
[700]??:91029902990299029D029D029D02A102A502A502A502A502A502A502A102A1029D0299029502950295029502990299029D02A102A102A502B102 [2026-04-22 13:16:48][DEBUG] ???????[301.59 ~346.56 ] [2026-04-22 13:16:48]
[DEBUG] [700]??:AA5501011 3AE13AE1656 [2026-04-22 13:16:48][DEBUG] ??? [2026-04-22 13:16:48][DEBUG] [700]??:BD02 [2026-04-22 13:16:48][DEBUG] ???????[348.14 ~348.14 ] [2026-04-22 13:16:48][DEBUG]
[700]??:AA550028CBAE1319EA72 [2026-04-22 13:16:48][DEBUG]
[700]??:C902D102DD02E902F502050322038603AE0300000000000000000000000000000000000000000000000000000000000A207000000009A97000000005A789A75620D660EB2240000000000000000 [2026-04-22 13:16:48]
[DEBUG] ???????[349.58 ~50.14 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA550028DB1919385308 [2026-04-22 13:16:48][DEBUG]
[700]??:000000000000D60A000000000000000000003E06CE060C0744078007B007B407DE084A063217FE0ABA08DA0A6A0C3E0AA6086D08160E2A08C6070000000096092A48FE0A0000000000000000CE040000 [2026-04-22 13:16:48]
[DEBUG] ???????[51.70 ~112.19 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA550028E5383357151C [2026-04-22 13:16:48][DEBUG]
[700]??:0000000000000000000000000000000000000001E0B0000AE0BE10B2C0C740C1E0A000000000000000000000000FA120000D2128A068D067C06E605160621063D066D06A106E2062107C60BA90BA10B [2026-04-22 13:16:48]
[DEBUG] ???????[113.78 ~174.39 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA550028FD576F76BF16 [2026-04-22 13:16:48][DEBUG]
[700]??:0000FE07000800000000000000B208C108CC08000000005A0B0000CA60CE5DBA5A3A58BA56A557D258D559D55A125CA25DF85CDE592A571254E250FE4D2E4B22516E4584440000000000000622161218222 [2026-04-22 13:16:48]
[DEBUG] ???????[175.97 ~236.86 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA5500283D77819565AF [2026-04-22 13:16:48][DEBUG]
[700]??:1523F92259222122112316240000000000A33E229F62A56316430E43030312204DE03AE0386036A034A032D0319030503F502E902D902D102C502BD02B502AD02A902A5029D029D029D029D02A5020000 [2026-04-22 13:16:48]
[DEBUG] ???????[238.47 ~299.00 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA55001F4D9681ADFC73 [2026-04-22 13:16:48][DEBUG]
[700]??:8A0295029902990299029D029D02A102A102A102A502A502A502A502A502A102A10299029502950295029502990299029D02A102A102A102A902B102 [2026-04-22 13:16:48][DEBUG] ???????[300.59 ~347.00 ] [2026-04-22 13:16:48]
[DEBUG] [700]??:AA5501014BAE4BAE6A56 [2026-04-22 13:16:48][DEBUG] ??? [2026-04-22 13:16:48][DEBUG] [700]??:C102 [2026-04-22 13:16:48][DEBUG] ???????[348.58 ~348.58 ] [2026-04-22 13:16:48][DEBUG]
[700]??:AA55002817AF5519EC0C [2026-04-22 13:16:48][DEBUG]
[700]??:CD02D502E102ED02F9020D0346039203000000000000000000000000000000000000000000000000000000000000000DE98000000006278 0000BA0D0000362B0000000000000000 [2026-04-22 13:16:48]
[DEBUG] ???????[350.17 ~50.66 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA5500281F1A5138E24A [2026-04-22 13:16:48][DEBUG]
[700]??:000000000000EE0A00000000000000000000006A06DE062A0758079A07B8070000E608C20A6E17060BBE082E0C3E0A040AA208D008220E1208C207000000000000005A0AFE0A0000000000000000CE040000 [2026-04-22 13:16:48]
[DEBUG] ???????[52.23 ~112.63 ] [2026-04-22 13:16:48][DEBUG] [700]??:AA5500281B397D574F32 [2026-04-22 13:16:48][DEBUG]
```

Deciphering LiDAR Packet (cont.)

- TTL output UART signal
- Convert TTL to RS-232 levels via MAX232 IC



Deciphering LiDAR Packet (cont.)

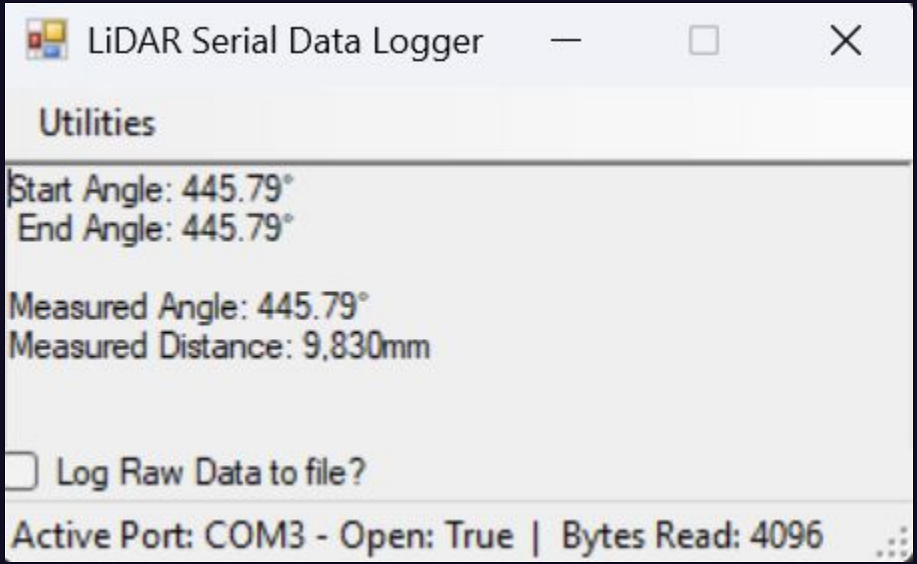
- C# program formatted file output
- Varying packet lengths
  - Unclear reasoning

```
ACQ: 20260430.1040:190 | DATA:
AA 55 00 28 4D 27 01 36 88 7C 28 38 F4 37 E1 37 30 37 99 37
6C 37 69 37 49 37 B2 2E 0A 32 D2 33 41 34 16 2F 4A 2D 59 2D
96 30 BA 31 72 35 72 30 E6 29 16 37 85 37 99 37 CD 37 7D 37
B4 37 00 00 00 00 A2 0E CE 09 A9 09 7D 09 65 09 49 09 2D 09
19 09 09 09 F9 08 E9 08 E5 08

ACQ: 20260430.1040:190 | DATA:
AA 55 00 28 59 36 0F 45 02 4C E5 08 ED 08 04 09 3A 0C 4D 0C
5D 0C 75 0C A1 0C CD 0C 09 0D 49 0D 95 0D BE 02 B9 02 BD 02
C5 02 C9 02 CD 02 D5 02 DD 02 E1 02 E9 02 F1 02 F9 02 FD 02
0D 03 00 00 00 00 00 00 CE 02 D9 02 5A 05 00 00 00 00 00 00
00 00 00 00 00 00 00 00 DA 42
```

Deciphering LiDAR Packet (cont.)

- Modified C# program attempts to convert UART data to readable values
- Conversion logic is obviously incorrect and/or broken
    - Code documentation already describes/indicates bug



Cost Analysis

Camera Board + Components	$54.52
Remote Boards	$255.00
Thermal Camera	$50,000.00
Sony Camera	$32,000.00
Misc Cameras	$70.00
LiDAR	$69.99
Rudy’s Time	232.5hrs @ $50 = $11,625.00
Jonas’s Time	63.0hrs @ $50 = $3,150
Total Cost	$97,224.51

Conclusions and Recommendations

- Camera
  - Redesign Camera PCB
  - Implement local human detection from video
  - Decode and utilize information sent back from the camera
  - Test Bloodhound system as a whole
  - Create TX and RX circuit
  - Create Video cleanup circuit
  - Find position hold command

- LiDAR
  - Different device with better documentation
  - Simplify/correct how to read and process data

Questions
?